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$$f(x) = (x^2 - 1)^4$$

$$f'(x) = 4(x^2 - 1)^3 \cdot 2x$$

$$= 8x(x^2 - 1)^3$$

$$f''(x) = 8(x(x^2 - 1)^3)'$$

$$= 8(1 \cdot (x^2 - 1)^3 + x \cdot 3(x^2 - 1)^2 \cdot 2x)$$

$$= 8(x^2 - 1)^3 + 48x^2(x^2 - 1)^2$$

$$= 8(x^6 - 3x^4 + 3x^2 - 1) + 48x^2(x^4 - 2x^2 + 1)$$

$$= 56x^6 - 120x^4 + 72x^2 - 8$$

$$f'''(x) = 336x^5 - 480x^3 + 144x$$

References